

Note: Write all the required equations (without simplification) to solve the following problems, then find the final answer.

Prob. 1: Consider the circuit shown in Fig. 1,

1. The open circuit voltage between $a-b$ terminals (V_o), when the load impedance is disconnected, is:

2. The short circuit current (I_{sc}), when $a-b$ terminals are shorted is:

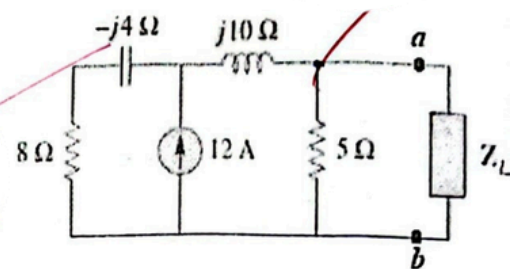


Fig.1

3. If the load impedance $Z_L = 4 + j3 \Omega$, The load current is:

Prob. 2: Consider the circuit shown in Fig. 2 and using mesh analysis method,

1. Write the two mesh equations, do not simplify:

2. The current I_1 is:

3. The current I_2 is:

4. The capacitor current, I_C , is: .

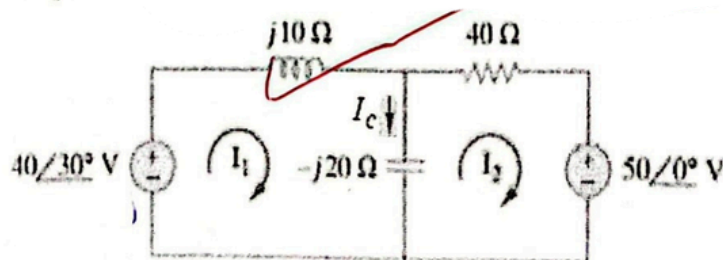


Fig 2

Prob.3:

Consider the circuit shown in figure 3, where a voltage source $v_s(t) = 50 \cos(500t + 15^\circ) \text{ V}$ supplies the load by a current $i(t) = 5 \cos(500t - 30^\circ) \text{ A}$

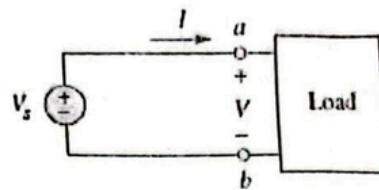


Fig.3

1. The equivalent impedance at the terminals of the voltage source is $Z = \quad \Omega$
2. The voltage lags the current by degrees.
3. The current leads the voltage by radians.
4. The voltage value $v_s(t)$ at $t = 2 \text{ ms}$ is V
5. The phasor domain representation of $v(t)$ is V
6. The frequency of the voltage $v_s(t)$ in Hz is Hz
7. The average value of the voltage $v_s(t)$ is V
8. The effective value of the current $i_s(t)$ is A
9. Draw the phasor diagram for the voltage $v_s(t)$ and the current $i_s(t)$.